

In W 17/03 the board was concerned with the investigation of technical relationships for unity. The case is summarised in the next section.

The boards have on several occasions stressed that an alleged lack of clarity in a claim cannot be used as a reason for an objection based on lack of unity (see W 31/88, OJ 1990, 134; W 7/89, W 59/90, W 9/02). In W 21/04 the board observed that according to the established case law the features of an independent claim that the ISA considered not to meet the requirements of Art. 6 PCT could not be disregarded in examining a lack of unity objection, as the IPEA had done here. See also W 8/07.

4.2. Examination as to novelty and inventive step

In decision G 1/89 (OJ 1991, 155; see also opinion G 2/89, OJ 1991, 166), the Enlarged Board of Appeal ruled on the question of the ISA's power to carry out a substantive examination of an international application in respect of novelty and inventive step when considering whether it complies with the requirement of unity of invention (see also referring decision W 12/89 (OJ 1990, 152).

If an International Searching Authority does have such power, in what circumstance does it have an obligation to carry out such a substantive examination?"

The Enlarged Board found that in accordance with normal terminology, the term "examination" or "substantive examination" referred to the activities of the authorities responsible for deciding on patentability, such as the examining division of the EPO, or the IPEA and/or the designated Office. Obviously, an ISA did not have any power to carry out such activities; it might only form a provisional opinion on novelty and inventive step for the purpose of carrying out an effective search. This opinion was in no way binding on the authorities referred to above. The same principle applied when an ISA considered that an international application did not comply with the requirement of unity of invention set forth in R. 13.1 PCT on an "a posteriori" basis. Such consideration had only the procedural effect of initiating the special procedure laid down in Art. 17 PCT and R. 40 PCT and was, therefore, not a "substantive examination" in the normal sense of that term. An assessment of whether a single general inventive concept existed was to be undertaken only in so far as this was necessary to establish whether the invention complied with the requirement of unity when implementing that procedure. Such an assessment was provisional and did not constitute a substantive examination in the sense of an examination as to patentability (see W 6/90, OJ 1991, 438).

The Enlarged Board further noted that consideration by an ISA of the requirement of unity of invention should always be made with a view to giving the applicant fair treatment and that the charging of additional fees under Art. 17(3)(a) PCT should be made only in clear cases. In particular, in view of the fact that such consideration under the PCT was being made without the applicant having had an opportunity to comment, the ISA should exercise restraint in the assessment of novelty and inventive step and in borderline cases preferably refrain from considering an application as not complying with the requirement of unity of invention on the ground of lack of novelty or inventive step (see W 24/90, W 23/91, W 43/91). In W 1/97 the board held that there was no such clear case where the ISA,

having already searched part of the subject-matter, was still not in a position to identify several separate inventions.

Following the Enlarged Board's conclusions in G 1/89 and G 2/89, the boards repeatedly used a lack of novelty or inventive step in the general inventive concept underlying the invention to justify a finding of lack of unity (see W 17/89, W 27/89, W 18/90 and W 19/90). In W 10/92 the board stressed that the problem-solution approach was to be applied to unity of invention (W 16/91, W 21/91).

In W 17/03 the ISA had considered that the absence of common technical features and an allegedly different problem solved would be sufficient to demonstrate lack of unity between groups of inventions. The board suggested that the ISA's approach to the determination of the problem appeared to stem from the belief that what was required was the same analysis as that used to determine the problem when using the problem and solution approach to evaluate inventive step, possibly because differences and problems are involved in both situations. The board stated that it was not convinced that the investigation of technical relationships for unity with the aid of common problems underlying the inventions is necessarily the same as the analysis that is used to determine the problem when using the problem and solution approach to evaluate inventive step. The board gave the following reasons:

(1) The evaluation of unity involves comparing problems solved (or effects achieved) by different claims, whereas the evaluation of inventive step is carried out on a single claim. As a result, when examining unity, the problems solved by different claims must be considered in the light of each other and cannot be determined in isolation in an absolute sense.

(2) In the evaluation of inventive step, the idea is to define a problem based on the distinguishing features that is essentially as narrow as possible, but not involving elements of the solution. On the other hand, in the evaluation of unity, these restrictions do not apply, since the overall object is to find out what the claims involved have in common, i.e. if the respective inventions are so linked as to form a single general inventive concept.

Thus, the specific problems solved by the different inventions with respect to the closest prior art might need gradual refinement, in particular generalisation starting from the problem directly solved, to find out whether or not there was a common denominator that still distinguished the inventions from the said prior art.

5. The single general inventive concept

5.1. General

When deciding on unity of invention, it is mandatory under Art. 82, R. 44 EPC and under R. 13.1 PCT to determine whether or not a group of inventions claimed in an application forms a single general inventive concept.

In W 19/89 the board decided that the application clearly lacked unity of invention, since the four possibilities comprised by claim 1 related to a further development of the state of the art in different directions, namely by employing different classes of dehalogenation agents having no new technical feature in common. Where at least one solution of the underlying technical problem already formed part of the state of the art, the requirement of a "single general inventive concept" implied that the further solutions to that problem proposed in the application must have at least one new element in common, this new element being normally represented by at least one new technical feature. Since the absence of such a unifying new technical feature was admitted by the applicant, the application related to more than one invention.

A useful analysis of the single general concept was made in W 6/90 (OJ 1991, 438). The board found that this concept manifested itself in features common to different teachings expounded individually in the same application. It observed that a teaching for the purposes of patent law encompassed not only the immediate subject-matter representing the solution to the problem as defined in the relevant claim, but also its technical consequences which were expressed as effects. It was noted that any subject-matter was defined by structural features and the relationship between them. The relevant effects, i.e. the outcome or results achieved by the invention as claimed, would usually already be apparent from the problem as stated. A single general concept might therefore be said to be present only if a partial identity existed between the teachings in an application deriving from the structural features of the subject-matters claimed and/or the outcome or results associated with those subject-matters.

Where subject-matters of the same category were concerned, a partial identity, generating unity of invention, could result from the structural features of these subject-matters and/or their associated effects. The absence of such an element common to all the different teachings in the application, and hence a lack of unity, might be established a priori under certain circumstances. A lack of unity might, however, also be established a posteriori between the subject-matters of different independent claims or in the remaining subject-matters if the subject-matter of a linking claim was clearly not novel or inventive vis-à-vis the state of the art. The board gave an example of what was meant by the abstract term "single general concept": a product, a process specially adapted for the manufacture of the said product, and a use of the said product, for example, embodied a single general concept because, on the one hand, the partial identity between the product and its use derived from the structural features of the product and, on the other hand, the partial identity shared by the product and the process specially adapted for its manufacture also derived from the product which was to be considered as the effect or result of this process (see T 119/82, OJ 1984, 217).

The board also noted that the criteria governing unity of invention for the purposes of R 13.1 PCT elucidated above also applied in principle where the inventive step was based chiefly on the discovery of an unrecognised problem (see T 2/83, OJ 1984, 265). If the common problem, i.e. the effects to be achieved, was itself already known or could be recognised as generally desirable (a mere desideratum) or obvious, there would be no inventive merit in formulating the problem. If the common structural features were to be

found only in the prior art portions of the claims, and if these known features did not help to solve the problem of the combined whole, this might also indicate a lack of unity.

In W 38/90 the sole common link between the subject-matter of claims 1, 2 and 4 was that the respective features all contributed to the realisation of the same device – a viewing assembly for a door. The board observed that this common pre-characterising feature was confined to indicating the designation of the invention, as part of the prior art; it was not, however, a specific feature interacting with the remaining characterising features and thereby contributing to the various inventions and their effects. Therefore, this sole link had to be rejected as irrelevant as regards any possible contribution to inventiveness. As already concluded, in the absence of any kind of common relevant feature, no unity "a priori" could be recognised.

In W 32/92 (OJ 1994, 239) the board held that there was no unity of invention where the subject-matters of independent claims, including their effects, had no common inventive feature in those parts of the claims differing from the closest prior art.

In T 861/92 the apparatus claimed was specifically adapted for carrying out only one step of the claimed process. The board came to the conclusion that the subject-matter of the claims satisfied the requirements of Art. 82 EPC 1973 as there was a technical relationship between both inventions.

In W 9/03 the board held that a common problem could establish the unity of different inventions only under certain conditions, for example in the case of a problem invention. According to the applicant, the common problem was considered that of providing a more secure, faster and easier connection setup. The board started from the assumption that the person skilled in the relevant art would normally consider this problem. The common problem was therefore so broadly formulated that it was "itself already known or could be recognised as generally desirable or obvious" (see W 6/90, OJ 1991, 438); as a result, unity could not thereby be established. See also T 2482/12.

5.2. Special technical features and inventive character of the single general concept

R. 13.2 PCT defines the method for determining whether the unity requirement is satisfied in respect of a group of inventions claimed in an international application: "Where a group of inventions is claimed in one and the same international application, the requirement of unity of invention referred to in R. 13.1 PCT shall be fulfilled only when there is a technical relationship among those inventions involving one or more of the same or corresponding special technical features. The expression "special technical features" shall mean those technical features that define a contribution which each of the claimed inventions, considered as a whole, makes over the prior art" (cf. R. 44(1) EPC).

According to the case law, once such a single, i.e. common, concept was established, it was necessary to consider whether or not the same could contribute to the inventiveness of the various subject-matters claimed in the case. Nothing indicated that this concept was known or belonged to the general knowledge of the man skilled in the art. Since the search

examiner did not indicate that the documents cited in the case could exclude such a contribution, it could not be assumed that this could not be the case (W 17/89; see W 6/90, OJ 1991, 438). In view of this, a single inventive concept had to be attributed to all claims linking all their subject-matters (see W 22/91).

In W 6/90 (OJ 1991, 438; see also in this chapter II.B.5.1. above) the board noted inter alia that R. 13.1 PCT also stipulated that the single general concept had to be inventive. Even with a given single general concept there was lack of unity if the concept had no inventive character. The boards of appeal have confirmed the above decision on several occasions (see e.g. W 31/91, W 29/92, W 34/92, W 45/92, W 8/93 and W 6/97).

In W 48/90 and W 50/90 the board noted that as far as chemical compounds were concerned, unity of invention was no mere question of the respective structural features, but had to be decided taking into account the technical problem to be solved and whether or not the respective compounds contributed to the solution found.

In W 45/92 the board stated that the term "inventive" could not be interpreted as a requirement for the common part to be inventive per se and therefore claimable as such. The investigation in this respect should concentrate on whether or not all or some of such features could contribute to the later detailed consideration of the inventive steps. Only if the state of the art or common general knowledge which had already been taught showed, beyond reasonable doubt, that this was not possible in the circumstances, should non-unity be declared. It further held that the term "same or corresponding special technical features" recognised that the features should define a contribution to the invention vis-à-vis the prior art. As the features in question were also part of the most relevant state of the art, where they were set out in an identical manner, they could not provide such a contribution. Thus, the various inventions, if any, had to lie in the unshared specific characterising features in each case. Since the claims contained no other common feature between them in the characterising part, this confirmed the conclusion of lack of unity (see W 32/92, OJ 1994, 239).

In W 38/92 the board confirmed W 6/90 (see above) and noted that the common features in the claim groups represented the "sole concept" linking the various subject-matters. The board stated that the remaining question was whether or not any of these specific features in the claim groups or their combination could make a contribution to the inventive steps relating to each claimed subject-matter; if so, this would render the said concept linking the various subject-matters inventive. The board observed that the new R. 13 PCT also required such a contribution by the suggested common "special technical features" over the prior art, i.e. that more than novelty be provided. In T 94/91 the board made it clear that the general inventive concept could not be equated with the features cited in a claim or in a particular combination of claims. What should be considered was the inventive concept as defined in the claims with due regard to the description and any drawings (see W 2/95).

In W 9/93 the board noted that the intermediate compounds were known, so that these products could not serve as a common inventive concept, and that a technical feature contained in the process claims was also known. The board pointed out that a technical